

Immigrant Neighborhood Formation: Evidence from Canadian Cities

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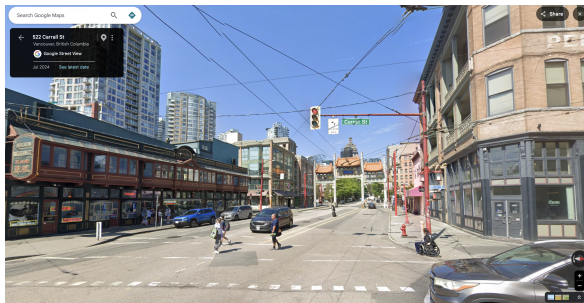
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Immigrant enclaves are not static

Vancouver Chinatown, 1970

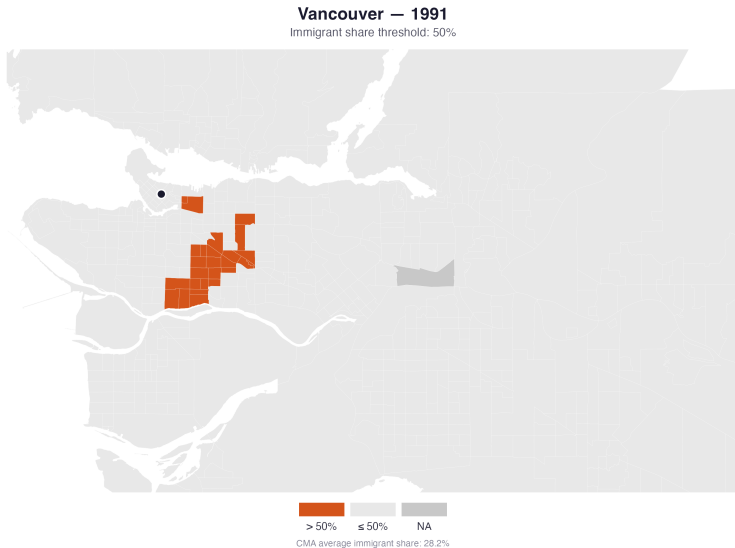


Vancouver Chinatown, 2024



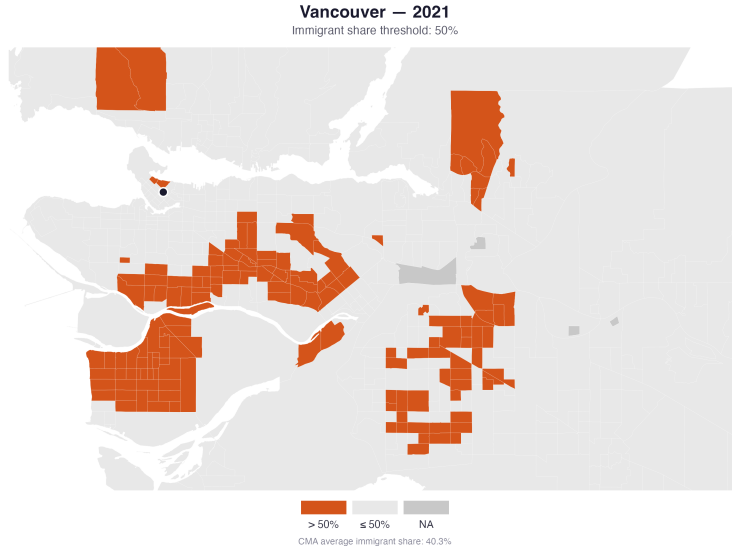
Vancouver's immigrant enclaves: 1991 → 2021

The geography of immigrant concentration has shifted fundamentally.



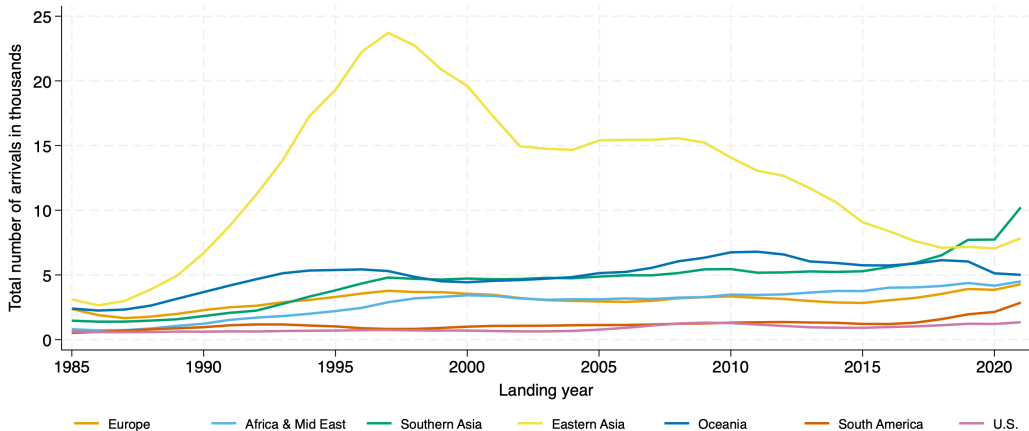
Vancouver's immigrant enclaves: 1991 → 2021

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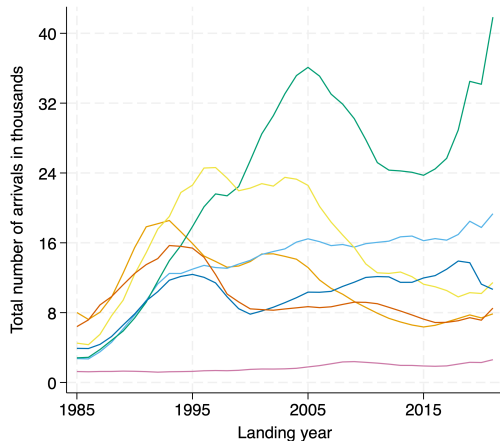
The shift in Chinatown coincides with dramatic changes in who arrives

Immigrant inflows to Vancouver, by region of origin (1985–2021).

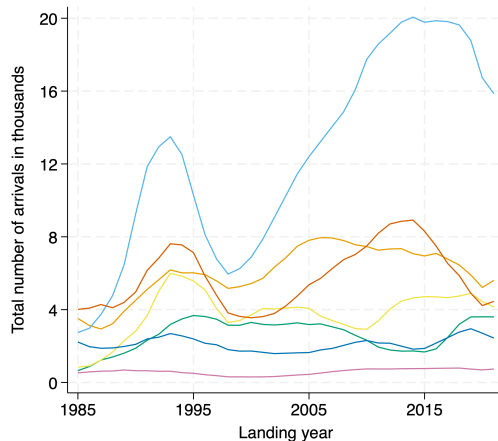


Compositional change in inflows is a national phenomenon

Toronto



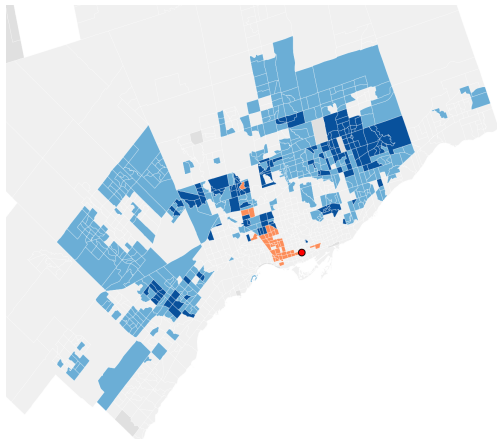
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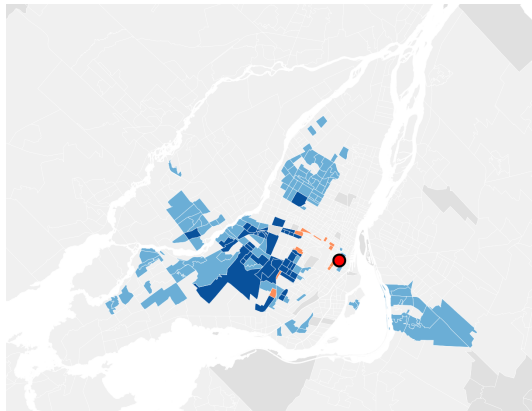
Europe Africa & Mid East Southern Asia Eastern Asia Oceania South America U.S.

Enclave evolution: Toronto and Montréal (1991 → 2021)

Toronto



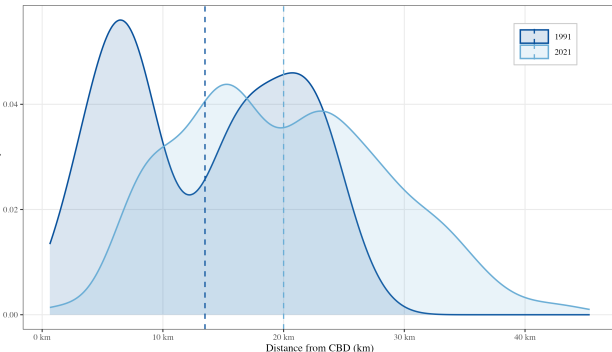
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Enclave status (1991 → 2021)



Enclaves are suburbanizing



Kernel density of enclave tracts by distance from CBD, pooling Montréal, Toronto, and Vancouver. Dashed lines = average distances.

Average enclave distance from CBD:

- 1991: ≈ 14 km
- 2021: ≈ 20 km

Enclaves are not just growing, they're moving outward.

Summary:

- Formation is ongoing
- Dissolution near CBDs
- Net suburbanization of ≈ 6 km over 30 years

Why do immigrant enclaves change over time?

Three candidate forces:

Preferences

Immigrants seek co-ethnic neighbors for social networks, language, amenities

Economic constraints

Budget limits restrict location choice; higher incomes allow integration

Inflow composition

Who arrives, and in what volumes, reshapes local demographic tipping points

Open questions:

- How much of enclave formation reflects **preferences** vs. **economic constraints**?
- How does this differ across immigrant groups — and relative to native-born?
- Can immigration *policy* shape enclave dynamics?

This paper

Study the determinants of immigrant enclave formation in Canadian cities.

Canadian Administrative Data

- ⇒ Universe of immigrants + 20% sample of Canadian-born taxpayers
- ⇒ Year-to-year 6-digit postal codes from 1985 to 2021 (LAD + IMDB)

A New Measure of Neighborhood Valuation

- ⇒ Adapt the **PageRank** algorithm to bilateral residential mobility flows
- ⇒ Microfound within a canonical Quantitative Spatial Model (QSM)

Dynamic Quantitative Urban Model

- ⇒ Forward-looking agents, endogenous amenities, elastic housing supply
- ⇒ Separate *preferences* from *prices* and *moving costs*

Counterfactual Simulations

- ⇒ Policy levers: immigration levels, income profiles, origin composition

Why do we care? Enclaves have large consequences

Enclaves shape immigrant outcomes.

- Enclave residence affects labor market integration, language acquisition, and social mobility (Munshi 2003; Edin et al. 2003; Damm 2009; Beaman 2012)
- Effects differ by income, education, and origin group

Enclaves are a cornerstone of immigration research design.

- Shift-share / enclave instruments widely used to estimate the effects of immigration on native outcomes (Altonji & Card 1991; Dustmann et al. 2013)
- If enclaves are endogenous to policy, so are the instruments

Policy relevance is direct.

- Canada has explicit tools to shape who arrives and where they settle
- Understanding the preference vs. constraint margin tells us which levers actually work

This paper is the first to jointly identify preferences, prices, and moving costs in a structural model of immigrant neighborhood formation.

Step 1: PageRank and neighborhood valuation — intuition

The revealed-preference idea. (Sorkin 2018)

- People move from neighborhoods they value less toward neighborhoods they value more
- A neighborhood is **highly valued** if it attracts residents from *other* highly valued places and retains them
- Bilateral flows encode all this information simultaneously

Why not hedonic regressions?

- Amenities are high-dimensional; no need to enumerate them
- We recover the *full indirect utility* of a location (amenities + labor market + housing cost), not just one attribute
- Works for *renters* as well as homeowners — critical for immigrants

Analogy: a web page is highly ranked if it receives links from other highly-ranked pages.

Step 1: PageRank fixed-point equation

Setup. For group k , let $\Upsilon_{h \rightarrow j}^k$ be the number of households moving from neighborhood h to j per year.

Key ratio. Under the logit model with symmetric within-city moving costs:

$$\frac{\Upsilon_{h \rightarrow j}^k}{\Upsilon_{j \rightarrow h}^k} \approx \frac{\tilde{V}^k(j)}{\tilde{V}^k(h)}, \quad \tilde{V}^k(j) \equiv \frac{e^{u^k(j) + \beta EV^k(j)}}{M_j^k}.$$

Cross-multiplying and summing over origins:

$$\tilde{V}^k(j) = \sum_{h \in \mathcal{J}} \frac{\Upsilon_{h \rightarrow j}^k \tilde{V}^k(h)}{\sum_h \Upsilon_{j \rightarrow h}^k}, \quad \forall j \in \mathcal{J}$$

This is the **PageRank fixed point**. $\tilde{V}^k(j)$ is the leading left eigenvector of the row-normalized bilateral flow matrix.

Microfounding PageRank in a Quantitative Spatial Model

Model structure.

- Type- k households choose location j each period, trading off flow utility, moving costs, and future expected values
- Indirect flow utility: $u_t^k(j) = \log b_{jt}^k + \log w_{kt} + (\varphi^k - 1) \log Q_{jt}$
- Amenities b_{jt}^k are **endogenous**: $\log b_{jt}^k = \Gamma_b^k M_{jt}$

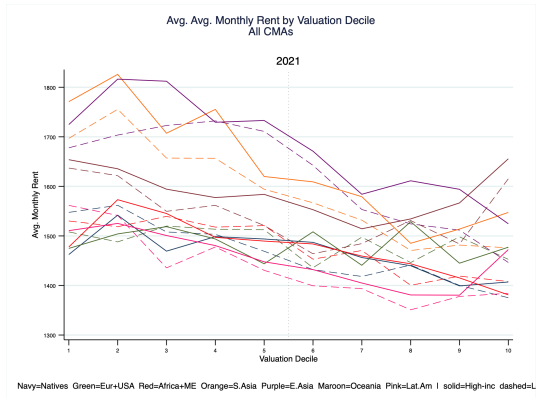
What PageRank identifies.

$$\tilde{V}^k(j) = \underbrace{e^{u^k(j)}}_{\text{flow utility}} \cdot \underbrace{e^{\beta EV^k(j)}}_{\text{discounted future value}} / M_j^k$$

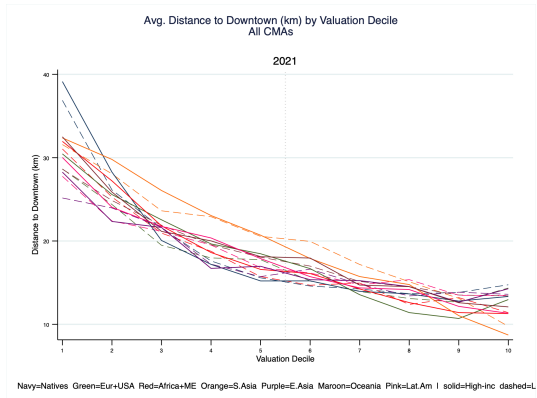
⇒ Bundles current *and* future neighborhood value — no distributional assumptions on amenities needed

Contribution: we are the first to microfound PageRank in a standard dynamic QSM, giving it a clean indirect-utility interpretation in the spatial economic context.

What does people value?



Valuation vs. Average Rent



Distance to Downtown vs. Valuation

Identification: Mismatched residents instrument

Idea: regress indirect utility over population composition to retrieve preference parameters

$$\tilde{V}_{jt}^k = \Gamma_b^k M_{jt} + \xi_{jt}^k$$

Problem: M_{jt} is endogenous!

Solution: Instrument using information that is orthogonal to the current valuation.

Mismatched residents (Caetano & Maheshri 2025): Individuals who picked their location based on amenities that are not relevant anymore, but are now sticking to their location because of moving costs.

$\Rightarrow M_{j,t-2}$ will predict M_{jt} , but might be correlated with ξ_{jt}^k . However, $M_{j,t-2} \perp \xi_{jt}^k | \tilde{V}_{j,t-1}^k$.

- **Relevant.** Stranded households still *physically contribute* to today's composition M_{jt} ; lagged composition $M_{j,t-2}$ predicts who is there now.
- **Valid.** The amenities that led them to sort there are *no longer relevant to new inflows*; once we control for last period's valuation $\tilde{V}_{j,t-1}^k$, past composition $M_{j,t-2}$ is orthogonal to today's unobservables ξ_{jt}^k .

Immigrant groups value co-ethnic neighbors — more so when poor

Immigrant groups strongly prefer their own. All immigrant groups show positive own-group coefficients

Low-income households value co-ethnic neighbors more. For 5 of 6 immigrant groups, the poor coefficient *exceeds* the rich coefficient on own-group share.

	Natives		Eur.+USA		Africa+ME		S. Asia		E. Asia		Oceania		Lat. Am.	
	Rich	Poor	Rich	Poor	Rich	Poor	Rich	Poor	Rich	Poor	Rich	Poor	Rich	Poor
<i>Eur.+USA Share</i>	-0.4718*** (0.1128)	-0.4256*** (0.1156)	0.5316*** (0.1188)	0.6790*** (0.1069)	-0.1957 (0.12)	0.1277 (0.1012)	0.2964** (0.1194)	0.018 (0.1138)	-0.1085 (0.1277)	0.0647 (0.1215)	0.0273 (0.12)	0.0252 (0.1211)	-0.2263* (0.1361)	-0.0881 (0.1046)
<i>Africa+ME Share</i>	-1.697 (0.1112)	0.3201*** (0.085)	-0.1677 (0.1156)	0.1653 (0.1187)	0.3822*** (0.0999)	0.6066*** (0.0939)	-0.0606 (0.1198)	-0.0191 (0.1051)	-0.2637** (0.1158)	0.0028 (0.1177)	0.091 (0.1115)	-0.0347 (0.1406)	-0.2288 (0.1112)	-0.1726 (0.1112)
<i>S. Asia Share</i>	0.0714 (0.0759)	0.2856*** (0.0642)	-0.0447 (0.0979)	-0.0082 (0.0879)	-0.0754 (0.0769)	-0.1203 (0.0744)	0.1021 (0.0654)	0.1043* (0.0608)	-0.0916 (0.0787)	-0.0936 (0.0807)	-0.0107 (0.0883)	-0.2313*** (0.078)	-0.3175*** (0.1372)	-0.1537** (0.0861)
<i>E. Asia Share</i>	0.0268 (0.0736)	0.2886*** (0.0532)	0.0226 (0.0825)	0.0745 (0.0727)	0.0125 (0.0588)	0.0901 (0.0525)	0.1034 (0.0711)	0.0295 (0.0654)	0.1397** (0.0658)	0.3115*** (0.0658)	0.0156 (0.069)	-0.2355*** (0.072)	-0.1372 (0.0861)	-0.1990*** (0.0743)
<i>Oceania Share</i>	0.0927 (0.1059)	0.1487* (0.0858)	0.1436 (0.123)	0.0712 (0.1059)	-0.1667 (0.1159)	-0.0616 (0.0958)	0.1474 (0.0984)	-0.0465 (0.1108)	0.1565 (0.1052)	0.165 (0.1121)	0.3466*** (0.1021)	0.5705*** (0.103)	0.2768** (0.1201)	0.1106 (0.1004)
<i>Lat. Am. Share</i>	0.0525 (0.1333)	0.2532** (0.1196)	0.3286* (0.1754)	-0.0785 (0.1766)	0.3770*** (0.1451)	0.0622 (0.1387)	-0.0005 (0.1509)	-0.116 (0.1433)	0.5148** (0.2095)	0.5548*** (0.1812)	-0.9272*** (0.1545)	-0.8862*** (0.155)	0.4407*** (0.1508)	0.3368*** (0.1385)
KP F-stat	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79	170.79
N	2650	2650	2585	2579	2635	2612	2534	2508	2541	2398	2617	2602	2583	2581

Dep. var.: PageRank neighborhood valuation percentile by group. Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Omitted group: natives; omitted income: poor.

Next steps

So far:

- ✓ PageRank neighborhood valuations estimated for all groups
- ✓ IV identification of co-ethnic response functions

In progress:

- **Moving costs** — estimate m_1^k, m_2^k via product-of-ratios (Conditional Choice Probability)
- **Dynamic model fit** — simulate the model forward from 1991; compare predicted vs. observed enclave patterns in 2021

Planned counterfactuals:

- Immigration *levels* ↑: how much more enclave formation?
- Richer intake: does income relax the housing constraint enough to change sorting?
- Origin composition shift: non-linear tipping dynamics under alternative inflow mixes